UNC Charlotte

ECGR 3120 Electromagnetics

Magnetically Steered Cardiac Ablation

Optional Extension — Coil Inductance and Transient Supply Voltage

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Input status: Coil radius, spacing, turns, 200 A baseline current, 162 mm^2 conductor area, and copper conductivity are from the official starter kit. Winding-pack shape, fill factor, and turn placement remain illustrative assumptions.

Abstract

This bonus extension estimates self- and mutual-inductance of the two navigation coils and the voltage required to command a 0–200 A current ramp in 100 ms. Unlike the earlier draft, the conductor area is now the official starter-kit value of 162 mm^2, rather than the previously assumed 4 mm conductor radius. Using an illustrative square winding pack with packing factor 0.65, the estimated series-aiding pair inductance is 40.303 mH, the peak prescribed-ramp terminal voltage is 95.116 V, and stored magnetic energy is 806.1 J. These are sensitivity-dependent engineering estimates, not measured coil properties.

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1 Objective and engineering context

The navigation field is proportional to current through a Helmholtz coil pair. Rapid steering requires a power supply capable of overcoming both copper resistance and induced voltage from changing magnetic flux. The additional extension estimates this transient voltage and the energy stored in the pair. It does not change the five required milestones.

2 Official inputs versus assumptions

Table 1: Starter-kit parameters and extension-only assumptions.

Parameter	Value	Basis
Mean coil radius and axial separation	0.350 m each	Starter-kit constants
Turns per coil	155	Starter-kit constant
Baseline current per coil	200 A	Starter-kit constant
Copper conductor area	162 mm^2	Starter-kit constant
Copper conductivity	5.8×10^7 S/m	Starter-kit constant (20 °C)
Connection	Series aiding	Same field orientation as M1
Current change	0 to 200 A in 100 ms	Selected bonus scenario
Packing factor	0.65	Assumed, not official
Pack shape	Square radial/axial section	Assumed, not official
Turn-center layout	13 by 12 rectangular lattice	Numerical approximation; 155 positions used

The official $162\,\text{mm}^2$ wire area corresponds to a circular-equivalent bare-wire radius of 7.181 mm; this is *not* a claim about actual wire shape. With a square cross-section and assumed fill factor $f = 0.65$, the pack side is

$$w = \sqrt{\frac{NA_{\text{Cu}}}{f}} = \sqrt{\frac{155(162 \times 10^{-6}\ \text{m}^2)}{0.65}} = 196.55\,\text{mm}.$$

The actual layer structure, insulation, conductor shape and pack dimensions are unknown. The nominal square pack is wide compared with the coil mean radius, so the inductance is particularly sensitive to the assumed winding geometry.

3 Inductance and circuit model

3.1 Self-inductance

For an air-core circular multilayer winding, Wheeler's empirical engineering formula gives

$$L_{\text{self}}[\mu\text{H}] \approx \frac{0.8 a_{\text{in}}^2 N^2}{6a_{\text{in}} + 9b_{\text{in}} + 10c_{\text{in}}},$$

where a is mean radius and b, c are axial and radial pack builds, all in inches. At the assumed pack size, $L_{\text{self}} = 15.888\,\text{mH}$ per coil. This approximation is not a measured inductance and is not independently certified for this broad pack geometry.

3.2 Mutual inductance

A pair of coaxial filament turns with radii a, b and axial separation z has mutual inductance

$$M_{\text{loop}} = \mu_0 \sqrt{ab} \left[\left(\frac{2}{k} - k \right) K(k^2) - \frac{2}{k} E(k^2) \right], \qquad k^2 = \frac{4ab}{(a+b)^2 + z^2},$$

with complete elliptic integrals K, E. We sum interactions between the assumed 155 turn centers in each coil, yielding $M \approx 4.264\,\text{mH}$. The assumed centers are distributed across the square pack; exact winding geometry may change this result.

For series-aiding coils,

$$L_{\text{pair}} = L_1 + L_2 + 2M = 2(L_{\text{self}} + M) \approx 40.303\,\text{mH}.$$

For series opposing, the mutual term changes sign; that is not the intended Helmholtz operating configuration.

3.3 Resistance, voltage, and magnetic energy

The starter-kit conductor cross-section sets the winding resistance using

$$\ell = N(2\pi a) = 340.863\,\text{m}, \qquad R_{\text{coil}} = \frac{\ell}{\sigma A_{\text{Cu}}} = 0.036\,277\,4\,\Omega, \qquad R_{\text{pair}} = 2R_{\text{coil}} = 0.072\,554\,9\,\Omega.$$

For an *imposed linear current ramp* $I(t) = 200t/0.1$, the slope is $dI/dt = 2000\,\text{A}\,\text{s}^{-1}$. Kirchhoff's voltage law gives

$$V(t) = L_{\text{pair}} \frac{dI}{dt} + R_{\text{pair}} I(t), \qquad W(t) = \tfrac{1}{2} L_{\text{pair}} I(t)^2.$$

This is a time-varying commanded voltage, not a constant-voltage RL step response.

4 Corrected numerical results

Table 2: Recomputed 100 ms ramp using the official conductor area.

Quantity	Result
Equivalent circular bare-wire radius (reference only)	7.181 mm
Assumed square winding-pack side	196.547 mm
Wheeler self-inductance, each coil	15.888 mH
Assumed-pack mutual inductance	4.264 mH
Total series-aiding inductance	40.303 mH
Wire length per coil	340.863 m
Resistance per coil	0.036277 Ω
Resistance, two coils in series	0.072555 Ω
Inductive voltage $L_{\text{pair}} dI/dt$	80.605 V
End-of-ramp resistive voltage $R_{\text{pair}} I_f$	14.511 V
Peak required terminal voltage	**95.116 V**
Magnetic stored energy at 200 A	806.053 J
Coil-pair DC copper loss at 200 A	2.902 kW

The voltage peaks at the end of the controlled linear ramp, just before the current slope becomes zero. During the ramp the inductive term is constant while RI rises. At 200 A, $W = \frac{1}{2}(0.0403026)(200)^2 \approx 806.1\,\text{J}$. This is stored energy, not additional steady-state copper heating.

5 Simulation figures

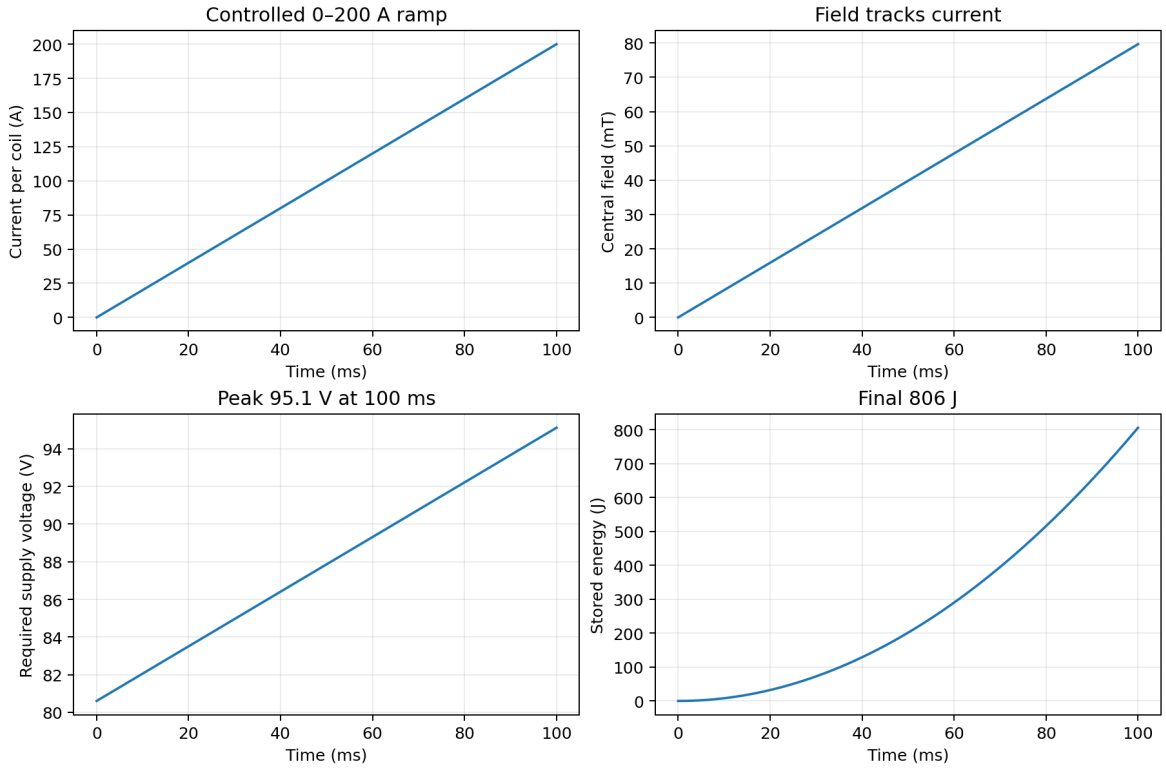


Figure 1: Corrected 0–200 A prescribed linear current ramp. All voltages and energy values are recalculated using 162 mm^2 copper area and an assumed winding pack.

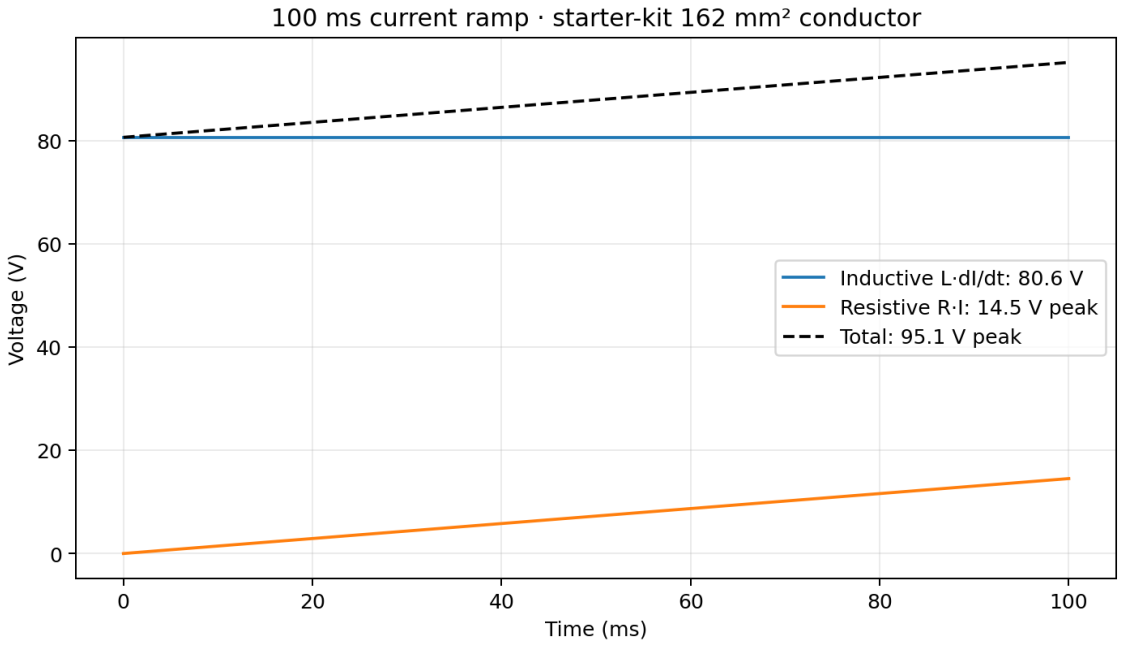


Figure 2: The required voltage is the sum of inductive and resistive components during the imposed 100 ms ramp. It peaks at 95.116 V.

6 Verification and assumption sensitivity

The corrected calculation checks $L_{\text{pair}} = 2L_{\text{self}} + 2M$, $V_{\text{total}} = V_L + V_R$ at every plotted time point, positive coupling, $W = \frac{1}{2}LI^2$, and consistency of I^2R with the official baseline coil loss. All these local checks passed during the corrected run. The older report's six local bonus unit-test results were for the *previous* wire-area scenario and are not presented here as a fresh test run. Instructor-specific bonus tests were not supplied.

Table 3: Sensitivity to winding fill and radial/axial pack aspect ratio.

Fill	w_r/w_a	L_{self} (mH)	M (mH)	L_{pair} (mH)	V_{peak} (V)
0.55	0.7	14.984	4.495	38.959	92.429
0.55	1.0	15.049	4.284	38.665	91.842
0.55	1.3	14.896	4.159	38.111	90.732
0.65	0.7	15.822	4.441	40.526	95.563
0.65	1.0	15.888	4.264	40.303	95.116
0.65	1.3	15.731	4.157	39.776	94.062
0.75	0.7	16.555	4.402	41.914	98.339
0.75	1.0	16.622	4.249	41.742	97.994
0.75	1.3	16.463	4.155	41.236	96.982

The selected geometry scenarios yield 38.111–41.914 mH inductance and 90.732–98.339 V peak voltage, with copper area held fixed at its official value. These are *scenario ranges*, not confidence bounds. Other layouts may lie outside them.

7 Engineering limitations and conclusion

The official 162 mm^2 conductor area corrects the resistance input and is not interchangeable with the prior 4 mm bare-wire radius. However, conductor area alone does not determine pack dimensions, insulation, turn arrangement or self-inductance. Wheeler's approximation, the 13×12 assumed turn-center layout, constant room-temperature copper conductivity, and ideal circular coils must be verified against any actual winding construction. Thermal changes, AC resistance, driver losses, protective flyback paths, and stored-energy recovery are outside this model.

Engineering takeaway. For the stated nominal pack assumption, about 95.1 V peak terminal voltage is needed to force a 100 ms 0–200 A ramp and about 0.806 kJ is stored at the endpoint. The power supply still must manage the approximately 2.90 kW resistive loss at 200 A and safely handle the stored energy.